

Engineering Physics: BAPHY105

Schrödinger Wave Equation

Dr. Samir Ranjan Meher

samirmeher@vit.ac.in, samirmeher@gmail.com

Department of Physics, VIT Vellore

Lecture Outcomes

1 Introduction

2 Formulation of Schrödinger wave equation

- Time dependent Schrödinger wave equation
- Time independent Schrödinger wave equation

Introduction

- ✓ Schrödinger (1926) started with de-Broglie's relationship between the momentum of a particle and wavelength associated with its matter waves.
- ✓ A rigorous mathematical theory known as Wave Mechanics was developed.
- ✓ Wave mechanics describes behavior of atomic systems and their interaction with other particles and electromagnetic radiation.

Formulation of Schrödinger wave equation

- Consider a particle of mass m moving with a momentum p .
- Let $\Psi(x, t)$ be the wave function of the matter wave associated with this moving particle.
- This complex wave function in general can be expressed as,

$$\Psi(x, t) = Ae^{i(kx - \omega t)}$$

- Total energy of the particle is given by,

$$E = \frac{p^2}{2m} + V(x, t)$$

- From the expression of the complex wave function,

$$\frac{\partial \Psi}{\partial t} = -i\omega \Psi(x, t); \quad \frac{\partial \Psi}{\partial x} = ik \Psi(x, t)$$

$$\Rightarrow \frac{\partial^2 \Psi}{\partial x^2} = -k^2 \Psi(x, t) = -\frac{p^2}{\hbar^2} \Psi(x, t)$$

- Multiplying $\Psi(x, t)$ to the expression for energy, E

$$E\Psi(x, t) = \frac{p^2}{2m} \Psi(x, t) + V(x, t)\Psi(x, t)$$

$$\Rightarrow E\Psi(x, t) = -\frac{\hbar^2}{2m} \nabla^2 \Psi(x, t) + V(x, t)\Psi(x, t) \quad (\because \vec{p} \equiv -i\hbar \vec{\nabla})$$

- Again,

$$E = h\nu = \hbar\omega \quad \Rightarrow E\Psi(x, t) = -\frac{1}{i\omega} \frac{\partial \Psi}{\partial t} \hbar\omega = i\hbar \frac{\partial}{\partial t} \Psi(x, t)$$

$$\therefore \left[-\frac{\hbar^2}{2m} \nabla^2 + V(x, t) \right] \Psi(x, t) = i\hbar \frac{\partial \Psi(x, t)}{\partial t}$$

→ Time dependent Schrödinger wave equation

- $\hat{H}$ is the **Hamiltonian operator** given by,

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(x, t) \right] = \hat{H}$$

$$\Rightarrow \hat{H}\Psi(x, t) = E\Psi(x, t)$$

- **Eigenvalue equation:**

- ✓ $\hat{H}$ – Eigen-operator
- ✓ $\Psi(x, t)$ – Eigen-function
- ✓ E – Eigen-value

Time independent Schrödinger wave equation

- Applying the **separation of variable** technique to time dependent Schrödinger wave equation,

$$\Psi(x, t) = \Phi(x)\chi(t)$$

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V(x, t) \right] \Phi(x)\chi(t) = i\hbar \frac{\partial}{\partial t} (\Phi(x)\chi(t))$$

$$\Rightarrow \frac{\left[-\frac{\hbar^2}{2m} \nabla^2 + V \right] \Phi(x)}{\Phi(x)} = \frac{1}{\chi(t)} i\hbar \frac{\partial \chi(t)}{\partial t} = \text{const.} = E$$

$$\therefore i\hbar \frac{\partial \chi(t)}{\partial t} = E\chi(t) \Rightarrow \chi(t) = C.e^{-\frac{iEt}{\hbar}}$$

- Similarly for the space part,

$$\therefore \left[-\frac{\hbar^2}{2m} \nabla^2 + V(x) \right] \Psi(x) = E \Psi(x)$$

→ Time independent Schrödinger wave equation